

like the LCA Tejas, HAL Advanced Medium Combat Aircraft, EADS CASA C-295 twin-turboprop tactical military transport aircraft production jointly by Airbus Defence and Space and Tata Advanced Systems, single-engine aircraft production by SAAB (Gripen) with a yet-to-be-chosen Indian OEM, FICV project, missile development programme, new aircraft carriers, submarines and Marine Counter Measure Vessels all point to an increased focus on electronics. Critical avionics, night-vision devices, missile electronics, electronic warfare (ground-based and airborne) systems, software-defined radios, robotics, radars (of all kinds), UAV electronic navigation systems, weapon sensors and air defence systems are the much publicised favourites.

According to experts, major demand-generating applications will be weapon control systems, avionics and digitalisation of displays. In addition, there is a demand for single-board computers, advanced PCB design technology, cables and harnesses to meet aerospace standards; a holistic security network and indigenously developed security devices; and innovative maintenance, repair and operational solutions to sustain and upgrade existing systems.

Offsets to play a key role

Seventy per cent of survey participants (Fig. 2) feel that offsets will be on the rise. However, emphasis will be mainly on embedded intelligence. So offsets will have to be defined in terms of service offerings rather than hardware, quite quickly.

The remaining 30 per cent of respondents raised concerns about the increase in threshold limit for offsets, under DPP 2016, to ₹ 20 billion (around \$305 million) from the earlier limit of ₹ 3 billion. This is in contrast to global trends in defence offset thresholds, which hover around \$5-\$15 million in many countries (including Israel, Malaysia, Turkey and UAE).

There are fears that this increased threshold limit may lead to a situation

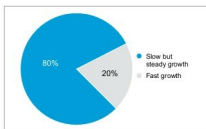


Fig. 1: Growth forecast for the Indian strategic electronics sector

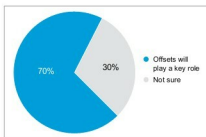


Fig. 2: Offsets-related opportunities

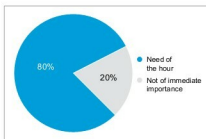


Fig. 3: Exploring design in India to achieve 'Make in India' goals

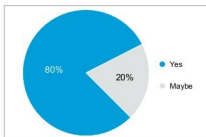


Fig. 4: Is public-private partnership a success?

where fewer import contracts will be eligible for offsets. This, in turn, will impact the domestic industry, particularly parts and components manufacturers who got the existing offset policy boost to enhance their capabilities.

Focus on 'design in India'

Survey participants feel that the Indian strategic electronics sector is still far from achieving self-reliance due to the absence of integration between

R&D and manufacturing. As many as eighty per cent (Fig. 3) respondents opined that this is the right time to explore 'Make in India' opportunities by designing in India for the country's defence electronics needs.

Last year, the defence sector got a policy boost through amendment of DPP 2016 to include a special category called 'Buy (Indian-Indigenously Designed, Developed and Manufactured)' or 'Buy (Indian-IDDM)'. This category will get the highest priority in the defence procurement process, followed by 'Buy and Make (Indian)', 'Buy and Make' and 'Buy (Global)' in that order. Inclusion of this new procurement category will provide a greater thrust to the 'Make in India' initiative in defence electronics production, focusing on products and solutions designed in India.

This would promote in-house design capabilities and higher localisation, strengthening the domestic industry, especially private sector, in defence electronics production. However, the responsibility to prove a design as indigenous rests with the manufacturer, while the final approval would come from the government. This, in turn, offers unique opportunities to companies that are operating or would like to enter the defence industry specifically in the areas of electronic system design and original design manufacturing.

The government has already launched some programmes and schemes to facilitate R&D activities in this field. The Technology Development Fund (TDF) and the Army Design Bureau (ADB) are worth mentioning here.

Partnering for success

According to survey participants, collaboration is the key to achieving self-reliance in the strategic electronics sector. There is a need for high standards in this industry, which can be fulfilled if the Indian industries complement each other rather than compete. The TDF and the ADB can play an